MODEL CONTEXT PROTOCOL
THE STANDARD THAT CONNECTS AI MODELS TO TOOLS, DATA, AND APPLICATIONS

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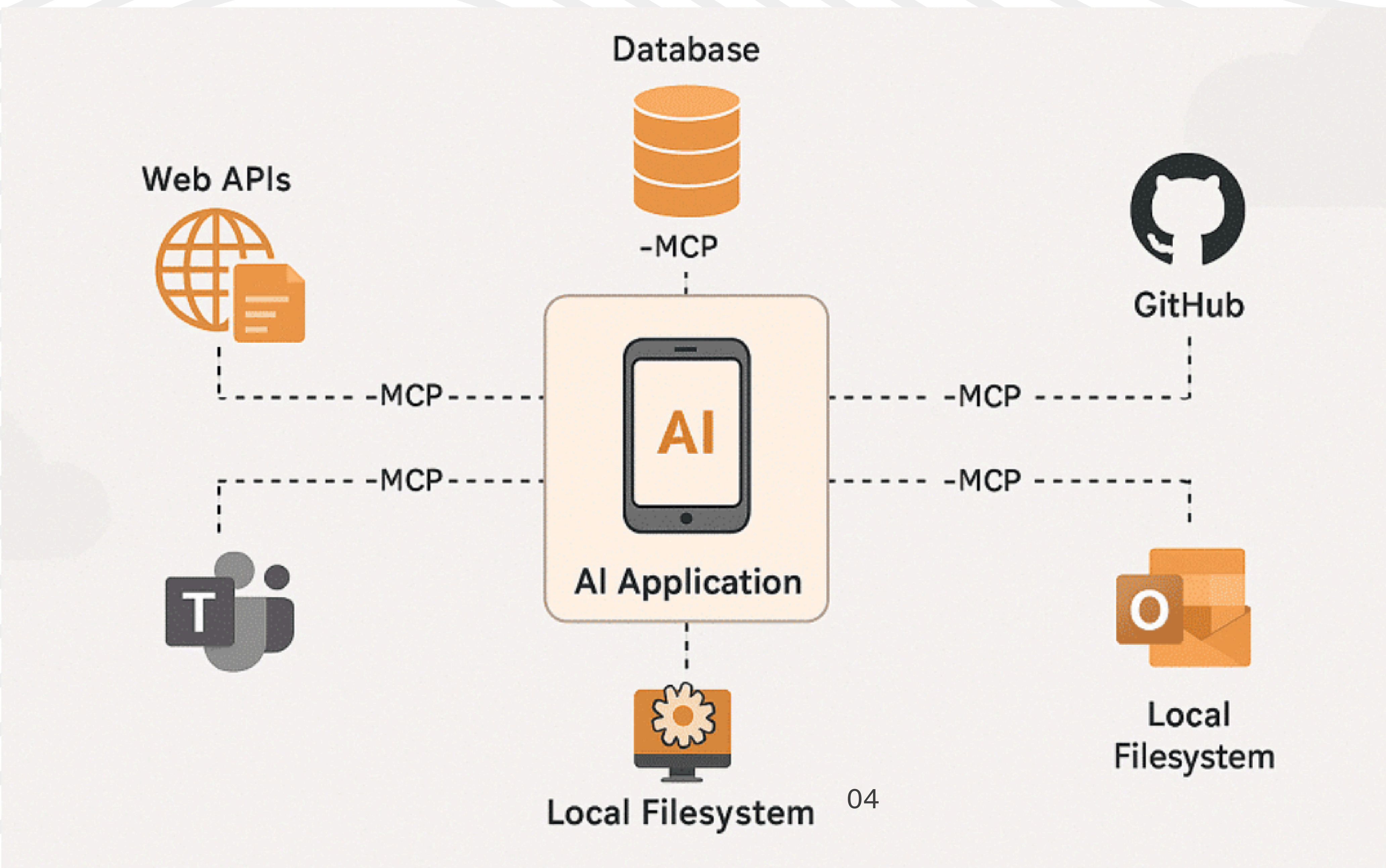
WHAT IS MCP

Model Context Protocol (MCP) is an open protocol that enables AI models to communicate with external systems through a common standard.

Think of MCP as USB-C for AI

- One standard
- Multiple tools
- Plug-and-play integrations
- Reduced complexity
- Better interoperability

MCP helps AI move beyond answering questions and start interacting with real-world systems.



WHY MCP WAS CREATED

Before MCP, every AI application needed separate integrations for every tool.

For example:

- One integration for PostgreSQL
- Another for Snowflake
- Another for GitHub
- Another for Slack
- Another for Google Drive

The Integration Problem

Integration complexity

High maintenance effort

Vendor-specific implementations

Duplicate engineering work

Slow development cycles

Build once, expose through MCP, and allow multiple AI systems to use the same interface.

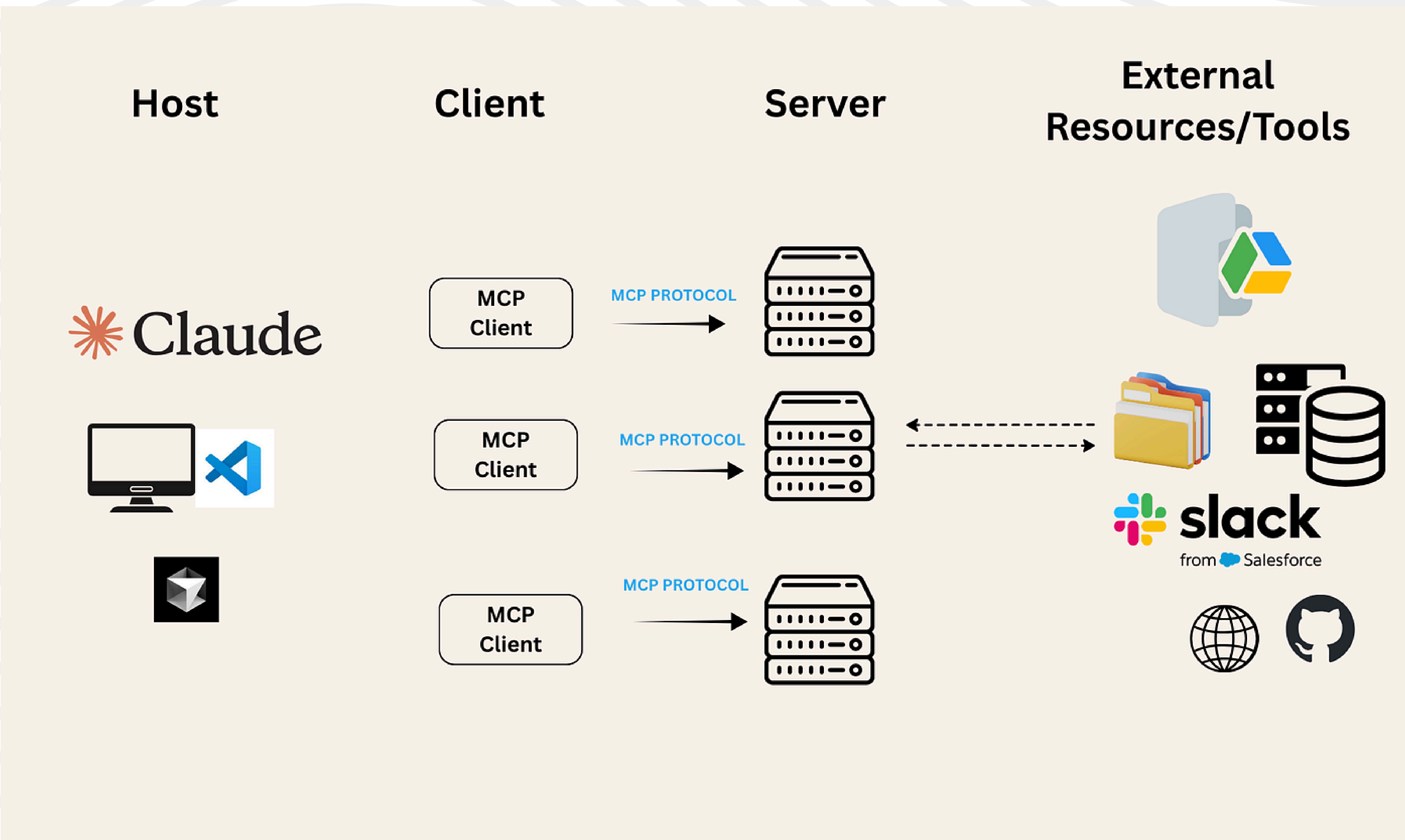
MCP ARCHITECTURE

MCP Host
The application where users interact with AI.

MCP Server
Provides access to tools, resources, and capabilities.

MCP Client
Responsible for communicating with MCP servers and handling requests.

External Systems
Includes Databases, APIs ,Cloud services , Data warehouses and Internal applications



HOW MCP WORKS

Step 1
User asks a question or requests an action.

Step 2
The AI identifies which tool or resource is needed.

Step 3
The MCP Client sends a request to an MCP Server.

Step 4
The MCP Server interacts with the target system.

Step 5
The result is returned to the AI model.

Step 6
The AI combines the information and responds to user.

Result
The model gains access to fresh, real-time information instead of relying only on training data.

MCP RESOURCES

Resources are pieces of information that AI can access for context.

📄 **Documentation**

📊 **Dashboards**

📁 **Files and folders**

🗄️ **Database schemas**

📈 **Reports**

🌐 **Knowledge bases**

Resources are identified using URIs that follow this format:

[protocol]://[host]/[path]

Text resources

- **Source code**
- **Configuration files**
- **Log files**
- **JSON/XML data**
- **Plain text**

Binary Resources

- **Images**
- **PDFs**
- **Audio files**
- **Video files**
- **Other non-text formats**

MCP TOOLS

Tools allow AI to perform actions instead of simply providing information.

Without tools:
AI only answers questions.
With tools:
AI can take actions and complete tasks.
This is what powers modern AI agents.

Examples

Execute SQL queries

Trigger Airflow workflows

Create Jira tickets

Send emails

Update databases

Call REST APIs

BENEFITS OF MCP

Faster Development

Reuse existing MCP integrations instead of building new ones.

Better Scalability

One integration can support multiple AI applications.

Improved Governance

Centralized control over tools and data access.

Reduced Maintenance

Fewer custom connectors to maintain.

Future-Proof Architecture

New AI models can connect using the same protocol.

Better Developer Experience

Consistent interfaces across systems.

KEY TAKEAWAY

The future of AI is not just better models.

The future is AI that can:

1. Access enterprise data
2. Understand business context
3. Use external tools
4. Execute workflows
5. Automate complex processes

MCP provides the foundation
that makes this possible.

THANK YOU